



Top 50 SQL Interview Coding Questions

Vol-1

Question-1

Write a query to fetch the car names where Kms_Driven is between 55000 and 78000, Price is less than 4 Lakhs, and Fuel_Type is CNG.

Table Script:

```
CREATE TABLE CAR_TB  
(Car_Name VARCHAR(20), Kms_Driven BIGINT, Price FLOAT, Fuel_Type VARCHAR(20));  
INSERT INTO CAR_TB VALUES  
( 'Maruti Alto',62000,3.6,'CNG'),  
( 'Hyundai i10',58000,3.8,'Petrol'),  
( 'Tata Nano',48000,2.1,'Petrol'),  
( 'Honda Amaze',73000,3.9,'CNG'),  
( 'Ford Figo',81000,4.2,'Diesel'),  
( 'Maruti WagonR',56000,3.5,'CNG'),  
( 'Toyota Etios',67000,3.95,'CNG');
```

Solution:

```
SELECT CAR_NAME  
FROM CAR_TB  
WHERE  
PRICE < 4  
AND  
FUEL_TYPE = 'CNG'  
AND  
KMS_DRIVEN BETWEEN 55000 AND 78000;
```

Answer:

Result Grid		Filter Rows:
	CAR_NAME	
▶	Maruti Alto	
	Honda Amaze	
	Maruti WagonR	
	Toyota Etios	

Question-2

Write an SQL query to find the second highest salary from the Employee table without using LIMIT or TOP?

Table Script:

```
CREATE TABLE EMP1(EMP_ID INT, EMP_NAME VARCHAR(20), SALARY BIGINT);  
INSERT INTO EMP1 VALUES  
(1,'Ram',89000),  
(2,'Boby',95000),  
(3,'Derek',70000),  
(4,'Diana',95000),  
(5,'Eve',60000),  
(6,'Kumar',92000);
```

Solution:

```
SELECT * FROM  
(SELECT *, DENSE_RANK() OVER(ORDER BY SALARY DESC)  
AS RNK FROM EMP1) A  
WHERE RNK = 2;
```

Answer:

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EMP_ID	EMP_NAME	SALARY	RNK
6	Kumar	92000	2

Question-3

Write SQL queries to retrieve only the odd-numbered and even-numbered rows from an Employee table?

Table Script:

```
CREATE TABLE EMP2 (EMPID INT, EMPNAME VARCHAR(20), DEPT VARCHAR(20), SALARY INT);
INSERT INTO EMP2 VALUES
(101,'Aakash','HR',50000),
(102,'Bhavana','Finance',60000),
(103,'Chandan','IT',70000),
(104,'Divya','Marketing',55000),
(105,'Eshan','IT',72000),
(106,'Farah','HR',52000),
(107,'Gaurav','Finance',61000),
(108,'Heena','IT',73000),
(109,'Ishaan','Marketing',54000);
```

Solution:

WITH CTE AS(

SELECT *, ROW_NUMBER() OVER(ORDER BY EMPID) AS RNK FROM EMP2)

SELECT * FROM CTE WHERE RNK%2=0;

WITH CTE AS(

SELECT *, ROW_NUMBER() OVER(ORDER BY EMPID) AS RNK FROM EMP2)

SELECT * FROM CTE WHERE RNK%2=1;

Answer:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	EMPID	EMPNAME	DEPT	SALARY	RNK
▶	102	Bhavana	Finance	60000	2
	104	Divya	Marketing	55000	4
	106	Farah	HR	52000	6
	108	Heena	IT	73000	8

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



EMPID	EMPNAME	DEPT	SALARY	RNK
101	Aakash	HR	50000	1
103	Chandan	IT	70000	3
105	Eshan	IT	72000	5
107	Gaurav	Finance	61000	7
109	Ishaan	Marketing	54000	9

Question-4

Write a SQL query(Single Query) to fetch the first and last record from a table without using where clause?

Table Script:

```
CREATE TABLE EMP3 (EMPID INT, EMPNAME VARCHAR(20), GENDER VARCHAR(10), LOCATION VARCHAR(20));
```

```
INSERT INTO EMP3 VALUES
```

```
(101,'Sheela','F','USA'),  
(102,'Rehan','M','USA'),  
(103,'Chandan','M','INDIA'),  
(104,'Divya','F','UK'),  
(105,'Nishant','M','CANADA'),  
(106,'Farah','F','UAE'),  
(107,'Gaurav','M','INDIA'),  
(108,'Heena','F','INDIA'),  
(109,'Ishaan','M','INDIA');
```

Solution:

```
(SELECT * FROM EMP3 ORDER BY EMPID ASC LIMIT 1)
```

```
UNION ALL
```

```
(SELECT * FROM EMP3 ORDER BY EMPID DESC LIMIT 1);
```

Answer:

EMPID	EMPNAME	GENDER	LOCATION
101	Sheela	F	USA
109	Ishaan	M	INDIA

Question-5

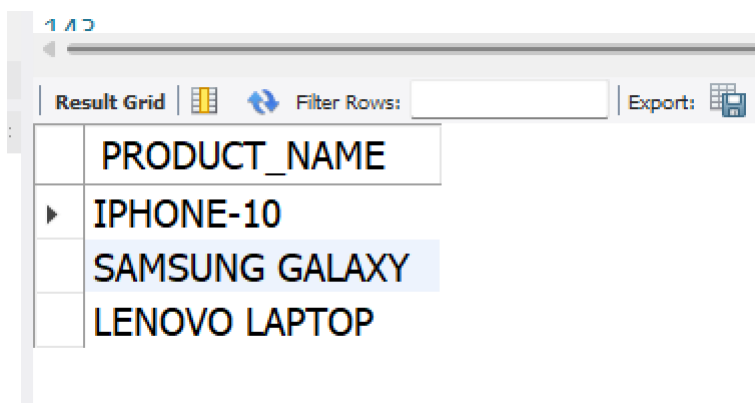
Write a query to fetch all products from the table whose price is greater than the average price of all products.

Table Script:

```
CREATE TABLE PRODUCT_TB1
(PRODUCT_ID TINYINT, PRODUCT_NAME VARCHAR(20), CATEGORY VARCHAR(20), PRICE INT);
INSERT INTO PRODUCT_TB1 VALUES
(1,"IPHONE-10","ELECTRONICS",79000),
(2,"SAMSUNG GALAXY","ELECTRONICS",67000),
(3,"LENOVO LAPTOP","ELECTRONICS",87999),
(4,"WATER BOTTLE","APPLINACES",299),
(5,"KEYBOARD","ELECTRONICS",999),
(6,"FIRE BOLT WATCH","ELECTRONICS",1999);
```

Solution:

```
SELECT PRODUCT_NAME FROM PRODUCT_TB1
WHERE PRICE > (
SELECT AVG(PRICE) FROM PRODUCT_TB1);
```

Answer:

	PRODUCT_NAME
▶	IPHONE-10
	SAMSUNG GALAXY
	LENOVO LAPTOP

Question-6

Write a Query to Show the Difference Between the Largest Weight and Smallest Weight for Patients with the Last Name 'Kumar'

Table Script:

```
CREATE TABLE PATIENT_TB
```

```
(patient_id VARCHAR(20), first_name VARCHAR(20), last_name VARCHAR(20), gender  
VARCHAR(20), birth_date DATE,
```

```
city VARCHAR(20), height INT, weight TINYINT);
```

```
INSERT INTO PATIENT_TB VALUES
```

```
(1,"Aarav","Sharma","M","1990-12-31","Delhi",168,51),
```

```
(2,"Siya","Kumar","F","1991-12-31","Mumbai",172,73),
```

```
(3,"Aryan","Singh","M","1992-12-31","Bangalore",160,93),
```

```
(4,"Diya","Gupta","M","1993-12-31","Chennai",160,79),
```

```
(5,"Rohan","Kumar","M","1994-12-31","Kolkata",173,87),
```

```
(6,"Ananya","Verma","F","1995-12-31","Hyderabad",185,51),
```

```
(7,"Vivaan","Kumar","M","1996-12-31","Pune",189,70),
```

```
(8,"Ishita","Sharma","M","1997-12-31","Ahmedabad",173,82),
```

```
(9,"Dhruv","Gupta","M","1998-12-31","Jaipur",152,61),
```

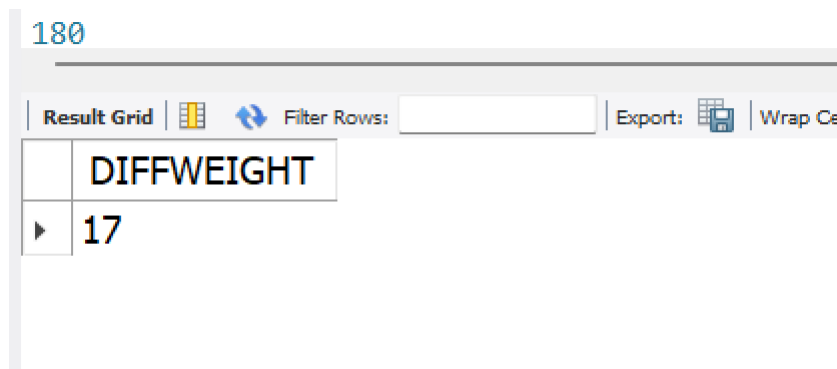
```
(10,"Myra","Singh","F","1999-12-31","Lucknow",171,71);
```

Solution:

```
SELECT MAX(WEIGHT) - MIN(WEIGHT) AS DIFFWEIGHT
```

```
FROM PATIENT_TB
```

```
WHERE LAST_NAME ="KUMAR";
```

Answer:

The screenshot shows a database query result in a grid format. The column header is 'DIFFWEIGHT' and the single data row contains the value '17'. The interface includes a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cells'.

DIFFWEIGHT
17

Question-7

Write a Query to Calculate the average weight for each gender, and also provide the count of patients for each gender and include patients who were born on or after January 1, 1994

Table Script:

```
CREATE TABLE PATIENT_TB
```

```
(patient_id VARCHAR(20), first_name VARCHAR(20), last_name VARCHAR(20), gender VARCHAR(20), birth_date DATE,
```

```
city VARCHAR(20), height INT, weight TINYINT);
```

```
INSERT INTO PATIENT_TB VALUES
```

```
(1,"Aarav","Sharma","M","1990-12-31","Delhi",168,51),
```

```
(2,"Siya","Kumar","F","1991-12-31","Mumbai",172,73),
```

```
(3,"Aryan","Singh","M","1992-12-31","Bangalore",160,93),
```

```
(4,"Diya","Gupta","M","1993-12-31","Chennai",160,79),
```

```
(5,"Rohan","Kumar","M","1994-12-31","Kolkata",173,87),
```

```
(6,"Ananya","Verma","F","1995-12-31","Hyderabad",185,51),
```

```
(7,"Vivaan","Kumar","M","1996-12-31","Pune",189,70),
```

```
(8,"Ishita","Sharma","M","1997-12-31","Ahmedabad",173,82),
```

```
(9,"Dhruv","Gupta","M","1998-12-31","Jaipur",152,61),
```

```
(10,"Myra","Singh","F","1999-12-31","Lucknow",171,71);
```

Solution:

```
SELECT
```

```
GENDER,
```

```
AVG(WEIGHT) AVGWEIGHT,
```

```
COUNT(GENDER) GCOUNT
```

```
FROM PATIENT_TB
```

```
WHERE BIRTH_DATE >= "1994-01-31" GROUP BY GENDER;
```

Answer:

Result Grid			
Filter Rows:		Export:	Wrap Cell Content
	GENDER	AVGWEIGHT	GCOUNT
▶	M	75.0000	4
	F	61.0000	2

Question-8

Find the patient_id, first_name, last_name, gender, and height of all patients who are taller than the average height of patients of their own gender. Order the results first by gender in ascending order, and then by height in descending order.

Table Script:

```
CREATE TABLE PATIENT_TB
```

```
(patient_id VARCHAR(20), first_name VARCHAR(20), last_name VARCHAR(20), gender VARCHAR(20), birth_date DATE,
```

```
city VARCHAR(20), height INT, weight TINYINT);
```

```
INSERT INTO PATIENT_TB VALUES
```

```
(1,"Aarav","Sharma","M","1990-12-31","Delhi",168,51),
```

```
(2,"Siya","Kumar","F","1991-12-31","Mumbai",172,73),
```

```
(3,"Aryan","Singh","M","1992-12-31","Bangalore",160,93),
```

```
(4,"Diya","Gupta","M","1993-12-31","Chennai",160,79),
```

```
(5,"Rohan","Kumar","M","1994-12-31","Kolkata",173,87),
```

```
(6,"Ananya","Verma","F","1995-12-31","Hyderabad",185,51),
```

```
(7,"Vivaan","Kumar","M","1996-12-31","Pune",189,70),
```

```
(8,"Ishita","Sharma","M","1997-12-31","Ahmedabad",173,82),
```

```
(9,"Dhruv","Gupta","M","1998-12-31","Jaipur",152,61),
```

```
(10,"Myra","Singh","F","1999-12-31","Lucknow",171,71);
```

Solution:

```
SELECT PATIENT_ID, FIRST_NAME, LAST_NAME, GENDER, HEIGHT
```

```
FROM PATIENT_TB P
```

```
WHERE
```

```
HEIGHT > (SELECT AVG(HEIGHT) FROM PATIENT_TB WHERE GENDER = P.GENDER)
```

```
ORDER BY GENDER ASC, HEIGHT DESC;
```

Answer:

Result Grid Filter Rows: Export: Wrap Cell Content:					
	PATIENT_ID	FIRST_NAME	LAST_NAME	GENDER	HEIGHT
▶	6	Ananya	Verma	F	185
	7	Vivaan	Kumar	M	189
	5	Rohan	Kumar	M	173
	8	Ishita	Sharma	M	173
	1	Aarav	Sharma	M	168

Question-9

Write a query to find the details of those salespeople who live in cities other than Pune and Delhi. Return broker_id, name, city, broker_commission.

Table Script:

```
CREATE TABLE BROKER_TB1
(Broker_Id INT,Name VARCHAR(20),City VARCHAR(20),Broker_Commission FLOAT);
INSERT INTO BROKER_TB1 VALUES
(101,"Alice Johnson","Mumbai",0.1),
(102,"Bob Williams","Pune",0.08),
(103,"Charlie Brown","Bangalore",0.12),
(104,"Diana Miller","Delhi",0.09),
(105,"Eve Davis","Chennai",0.11),
(106,"Frank White","Hyderabad",0.07),
(107,"Grace Taylor","Pune",0.08),
(108,"Henry Wilson","Kolkata",0.1),
(109,"Ivy Moore","Delhi",0.09),
(110,"Jack Green","Ahmedabad",0.12);
```

Solution:

```
SELECT Broker_Id,Name,City,Broker_Commission
FROM
BROKER_TB1
WHERE City NOT IN ("Pune","Delhi");
```

Answer:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Broker_Id	Name	City	Broker_Commission
101	Alice Johnson	Mumbai	0.1
103	Charlie Brown	Bangalore	0.12
105	Eve Davis	Chennai	0.11
106	Frank White	Hyderabad	0.07
108	Henry Wilson	Kolkata	0.1
110	Jack Green	Ahmedabad	0.12

Question-10

Write a SQL query to find the cities where the average broker_commission of salespeople is greater than 0.09. List these cities along with their average commission, ordered by the average commission in descending order.

Table Script:

```
CREATE TABLE BROKER_TB1
(Broker_Id INT,Name VARCHAR(20),City VARCHAR(20),Broker_Commission FLOAT);
INSERT INTO BROKER_TB1 VALUES
(101,"Alice Johnson","Mumbai",0.1),
(102,"Bob Williams","Pune",0.08),
(103,"Charlie Brown","Bangalore",0.12),
(104,"Diana Miller","Delhi",0.09),
(105,"Eve Davis","Chennai",0.11),
(106,"Frank White","Hyderabad",0.07),
(107,"Grace Taylor","Pune",0.08),
(108,"Henry Wilson","Kolkata",0.1),
(109,"Ivy Moore","Delhi",0.09),
(110,"Jack Green","Ahmedabad",0.12);
```

Solution:

```
SELECT City, AVG(Broker_Commission) AS AVGCOMMISSION
FROM BROKER_TB1
GROUP BY City
HAVING AVG(Broker_Commission) > 0.09
ORDER BY AVGCOMMISSION DESC;
```

Answer:

City	AVGCOMMISSION
Bangalore	0.11999999731779099
Ahmedabad	0.11999999731779099
Chennai	0.10999999940395355
Mumbai	0.10000000149011612
Kolkata	0.10000000149011612
Delhi	0.09000000357627869

Question-11

Write a Query to find their total purchase amount across all their orders for each customer. Return the customer's name and their total purchase amount. Only include customers whose total purchase amount is greater than 4000.

Table Script:

Table1:

```
CREATE TABLE CUST_TB1  
(CUST_ID INT, CUST_NAME VARCHAR(20));
```

```
INSERT INTO CUST_TB1 VALUES
```

```
(101,"John Doe"),  
(102,"Jane Smith"),  
(103,"Peter Jones"),  
(104,"Alice Brown"),  
(105,"Bob White");
```

Table2:

```
CREATE TABLE ORDERS_TB1  
(ORDER_ID INT,CUST_ID INT,ORDER_DATE DATE, PURCHASE_AMOUNT INT);
```

```
INSERT INTO ORDERS_TB1 VALUES
```

```
(1,101,"2023-01-10",1500),  
(2,102,"2023-01-10",2500),  
(3,101,"2023-01-10",2100),  
(4,103,"2023-01-11",1800),  
(5,102,"2023-01-11",3000),  
(6,101,"2023-01-12",2800),  
(7,104,"2023-01-12",1950),  
(8,105,"2023-01-13",3500),  
(9,101,"2023-01-10",1900),  
(10,102,"2023-01-10",1700),  
(11,103,"2023-01-11",2200),  
(12,104,"2023-01-14",2050),  
(13,105,"2023-01-13",1000);
```

Solution:

```
SELECT C.CUST_NAME,SUM(O.PURCHASE_AMOUNT) AS PURAMOUNT  
FROM CUST_TB1 C INNER JOIN ORDERS_TB1 O  
ON C.CUST_ID = O.CUST_ID  
GROUP BY C.CUST_NAME  
HAVING SUM(O.PURCHASE_AMOUNT) > 4000;
```

Answer:

Result Grid		Filter Rows:	Expo
	CUST_NAME	PURAMOUNT	
▶	John Doe	8300	
	Jane Smith	7200	
	Bob White	4500	

Question-12

Write a Query to find the average price of products for each product category.

Table Script:

```
CREATE TABLE PRODUCT_TB2
```

```
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(20), CATEGORY VARCHAR(20), SUB_CATEGORY VARCHAR(20),
```

```
PRICE FLOAT, STOCK_QTY INT);
```

```
INSERT INTO PRODUCT_TB2 VALUES
```

```
(101,"Laptop","Electronics","Computers",1200,25),
```

```
(102,"Smartphone","Electronics","Mobile",800,45),
```

```
(103,"Coffee Maker","Home Appliances","Kitchen",75.5,15),
```

```
(104,"T-Shirt","Apparel","Clothing",25,150),
```

```
(105,"Running Shoes","Footwear","Sports",95.75,80),
```

```
(106,"Bluetooth Speaker","Electronics","Audio",49.99,60),
```

```
(107,"Office Chair","Furniture","Office",250,12),
```

```
(108,"Novel","Books","Fiction",15.99,120);
```

Solution:

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```
SELECT C.CUST_NAME,SUM(O.PURCHASE_AMOUNT) AS PURAMOUNT
FROM CUST_TB1 C INNER JOIN ORDERS_TB1 O
ON C.CUST_ID = O.CUST_ID
GROUP BY C.CUST_NAME
HAVING SUM(O.PURCHASE_AMOUNT) > 4000;
```

Answer:

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
CATEGORY	AVGPRICE			
▶ Electronics	683.3300005594889			
Home Appliances	75.5			
Apparel	25			
Footwear	95.75			
Furniture	250			
Books	15.989999771118164			

Question-13

Write a Query to list all product categories and the number of products in each category

Table Script:

```
CREATE TABLE PRODUCT_TB2
```

```
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(20), CATEGORY VARCHAR(20), SUB_CATEGORY VARCHAR(20),
```

```
PRICE FLOAT, STOCK_QTY INT);
```

```
INSERT INTO PRODUCT_TB2 VALUES
```

```
(101,"Laptop","Electronics","Computers",1200,25),
```

```
(102,"Smartphone","Electronics","Mobile",800,45),
```

```
(103,"Coffee Maker","Home Appliances","Kitchen",75.5,15),
```

```
(104,"T-Shirt","Apparel","Clothing",25,150),
```

```
(105,"Running Shoes","Footwear","Sports",95.75,80),
```

```
(106,"Bluetooth Speaker","Electronics","Audio",49.99,60),
```

```
(107,"Office Chair","Furniture","Office",250,12),
```

```
(108,"Novel","Books","Fiction",15.99,120);
```

Solution:

```
SELECT CATEGORY,
```

```
COUNT(DISTINCT PRODUCT_NAME) AS PRODU_COUNT
```

```
FROM PRODUCT_TB2 GROUP BY CATEGORY;
```

Answer:

Result Grid		Filter Rows:	Export:	Wrap Ce
	CATEGORY	PRODU_COUNT		
▶	Apparel	1		
	Books	1		
	Electronics	3		
	Footwear	1		
	Furniture	1		
	Home Appliances	1		

Question-14

Write a Query to find the total value of stock for each product category.

Table Script:

```
CREATE TABLE PRODUCT_TB2
```

```
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(20), CATEGORY VARCHAR(20), SUB_CATEGORY  
VARCHAR(20),
```

```
PRICE FLOAT, STOCK_QTY INT);
```

```
INSERT INTO PRODUCT_TB2 VALUES
```

```
(101,"Laptop","Electronics","Computers",1200,25),
```

```
(102,"Smartphone","Electronics","Mobile",800,45),
```

```
(103,"Coffee Maker","Home Appliances","Kitchen",75.5,15),
```

```
(104,"T-Shirt","Apparel","Clothing",25,150),
```

```
(105,"Running Shoes","Footwear","Sports",95.75,80),
```

```
(106,"Bluetooth Speaker","Electronics","Audio",49.99,60),
```

```
(107,"Office Chair","Furniture","Office",250,12),
```

```
(108,"Novel","Books","Fiction",15.99,120);
```

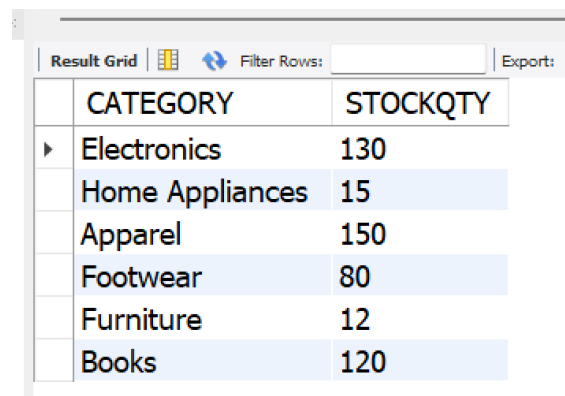
Solution:

```
SELECT CATEGORY,SUM(STOCK_QTY) AS STOCKQTY
```

```
FROM PRODUCT_TB2
```

```
GROUP BY
```

```
CATEGORY;
```

Answer:

CATEGORY	STOCKQTY
Electronics	130
Home Appliances	15
Apparel	150
Footwear	80
Furniture	12
Books	120

Question-15

Write a Query to find Second Highest Order by Each Customer**Table Script:**

```
CREATE TABLE ORDERS_TB2
```

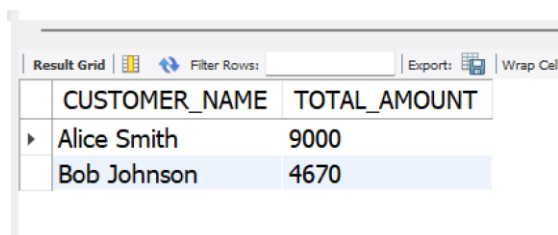
```
(ORDERID INT, ORDERDATE DATE, CUSTOMER_NAME VARCHAR(20), PRODUCT_NAME  
VARCHAR(30), ORDER_QUANTITY TINYINT, TOTAL_AMOUNT INT, PRODUCT_CATEGORY  
VARCHAR(20));
```

```
INSERT INTO ORDERS_TB2 VALUES
```

```
(1001,"2025-01-15","Alice Smith","Laptop Pro",1,57000,"Electronics"),  
(1002,"2025-01-15","Bob Johnson","Ergonomic Chair",1,4670,"Furniture"),  
(1003,"2025-01-16","Charlie Brown","Wireless Mouse",2,430,"Electronics"),  
(1004,"2025-01-17","David Lee","Coffee Maker",3,789,"Home Appliances"),  
(1005,"2025-01-18","Alice Smith","External Hard Drive",1,9000,"Electronics"),  
(1006,"2025-01-18","Eve Davis","Bluetooth Speaker",4,1200,"Electronics"),  
(1007,"2025-01-19","Frank Green","Desk Lamp",1,300,"Home Decor"),  
(1008,"2025-01-20","Grace Hall","Gaming Keyboard",2,560,"Electronics"),  
(1009,"2025-01-20","Bob Johnson","Smart TV",1,8000,"Electronics"),  
(1010,"2025-01-21","Ivy White","Air Fryer",1,1999,"Home Appliances");
```

Solution:

```
SELECT CUSTOMER_NAME, TOTAL_AMOUNT FROM (  
SELECT CUSTOMER_NAME, TOTAL_AMOUNT,  
DENSE_RANK() OVER(PARTITION BY CUSTOMER_NAME ORDER BY TOTAL_AMOUNT DESC) AS  
RNK  
FROM ORDERS_TB2) A  
WHERE RNK = 2;
```

Answer:

	CUSTOMER_NAME	TOTAL_AMOUNT
▶	Alice Smith	9000
	Bob Johnson	4670

Question-16

Write a SQL Query to find Cumulative Total Amount per Customer Over Time**Table Script:**

```
CREATE TABLE ORDERS_TB2
```

```
(ORDERID INT, ORDERDATE DATE, CUSTOMER_NAME VARCHAR(20),PRODUCT_NAME  
VARCHAR(30), ORDER_QUANTITY TINYINT, TOTAL_AMOUNT INT, PRODUCT_CATEGORY  
VARCHAR(20));
```

```
INSERT INTO ORDERS_TB2 VALUES
```

```
(1001,"2025-01-15","Alice Smith","Laptop Pro",1,57000,"Electronics"),  
(1002,"2025-01-15","Bob Johnson","Ergonomic Chair",1,4670,"Furniture"),  
(1003,"2025-01-16","Charlie Brown","Wireless Mouse",2,430,"Electronics"),  
(1004,"2025-01-17","David Lee","Coffee Maker",3,789,"Home Appliances"),  
(1005,"2025-01-18","Alice Smith","External Hard Drive",1,9000,"Electronics"),  
(1006,"2025-01-18","Eve Davis","Bluetooth Speaker",4,1200,"Electronics"),  
(1007,"2025-01-19","Frank Green","Desk Lamp",1,300,"Home Decor"),  
(1008,"2025-01-20","Grace Hall","Gaming Keyboard",2,560,"Electronics"),  
(1009,"2025-01-20","Bob Johnson","Smart TV",1,8000,"Electronics"),  
(1010,"2025-01-21","Ivy White","Air Fryer",1,1999,"Home Appliances");
```

Solution:

```
SELECT ORDERID, ORDERDATE, CUSTOMER_NAME, TOTAL_AMOUNT,  
SUM(TOTAL_AMOUNT) OVER(PARTITION BY CUSTOMER_NAME ORDER BY  
ORDERDATE,ORDERID) AS CUMULATIVE_TTL  
FROM ORDERS_TB2  
ORDER BY CUSTOMER_NAME,ORDERDATE,ORDERID;
```

Answer:

339

ORDERID	ORDERDATE	CUSTOMER_NAME	TOTAL_AMOUNT	CUMULATIVE_TTL
1001	2025-01-15	Alice Smith	57000	57000
1005	2025-01-18	Alice Smith	9000	66000
1002	2025-01-15	Bob Johnson	4670	4670
1009	2025-01-20	Bob Johnson	8000	12670
1003	2025-01-16	Charlie Brown	430	430
1004	2025-01-17	David Lee	789	789

Result 8 x

Question-17

Write a SQL Query to find Customers Who Ordered from at Least Two Distinct Product Categories**Table Script:**

```
CREATE TABLE ORDERS_TB2
```

```
(ORDERID INT, ORDERDATE DATE, CUSTOMER_NAME VARCHAR(20), PRODUCT_NAME  
VARCHAR(30), ORDER_QUANTITY TINYINT, TOTAL_AMOUNT INT, PRODUCT_CATEGORY  
VARCHAR(20));
```

```
INSERT INTO ORDERS_TB2 VALUES
```

```
(1001,"2025-01-15","Alice Smith","Laptop Pro",1,57000,"Electronics"),  
(1002,"2025-01-15","Bob Johnson","Ergonomic Chair",1,4670,"Furniture"),  
(1003,"2025-01-16","Charlie Brown","Wireless Mouse",2,430,"Electronics"),  
(1004,"2025-01-17","David Lee","Coffee Maker",3,789,"Home Appliances"),  
(1005,"2025-01-18","Alice Smith","External Hard Drive",1,9000,"Electronics"),  
(1006,"2025-01-18","Eve Davis","Bluetooth Speaker",4,1200,"Electronics"),  
(1007,"2025-01-19","Frank Green","Desk Lamp",1,300,"Home Decor"),  
(1008,"2025-01-20","Grace Hall","Gaming Keyboard",2,560,"Electronics"),  
(1009,"2025-01-20","Bob Johnson","Smart TV",1,8000,"Electronics"),  
(1010,"2025-01-21","Ivy White","Air Fryer",1,1999,"Home Appliances");
```

Solution:

```
SELECT CUSTOMER_NAME, COUNT(DISTINCT PRODUCT_CATEGORY) CNT  
FROM ORDERS_TB2  
GROUP BY CUSTOMER_NAME  
HAVING CNT >= 2
```

Answer:

Result Grid	Filter Rows:	Export:
CUSTOMER_NAME	CNT	
▶ Bob Johnson	2	

Question-18

Write a SQL Query to find Percentage Contribution of Each Order to Total Revenue

Table Script:

```
CREATE TABLE ORDERS_TB2
```

```
(ORDERID INT, ORDERDATE DATE, CUSTOMER_NAME VARCHAR(20),PRODUCT_NAME  
VARCHAR(30), ORDER_QUANTITY TINYINT, TOTAL_AMOUNT INT, PRODUCT_CATEGORY  
VARCHAR(20));
```

```
INSERT INTO ORDERS_TB2 VALUES
```

```
(1001,"2025-01-15","Alice Smith","Laptop Pro",1,57000,"Electronics"),  
(1002,"2025-01-15","Bob Johnson","Ergonomic Chair",1,4670,"Furniture"),  
(1003,"2025-01-16","Charlie Brown","Wireless Mouse",2,430,"Electronics"),  
(1004,"2025-01-17","David Lee","Coffee Maker",3,789,"Home Appliances"),  
(1005,"2025-01-18","Alice Smith","External Hard Drive",1,9000,"Electronics"),  
(1006,"2025-01-18","Eve Davis","Bluetooth Speaker",4,1200,"Electronics"),  
(1007,"2025-01-19","Frank Green","Desk Lamp",1,300,"Home Decor"),  
(1008,"2025-01-20","Grace Hall","Gaming Keyboard",2,560,"Electronics"),  
(1009,"2025-01-20","Bob Johnson","Smart TV",1,8000,"Electronics"),  
(1010,"2025-01-21","Ivy White","Air Fryer",1,1999,"Home Appliances");
```

Solution:

```
SELECT ORDERID,TOTAL_AMOUNT,  
(TOTAL_AMOUNT / SUM(TOTAL_AMOUNT) OVER()) * 100 AS PECNTGCONTRI  
FROM ORDERS_TB2  
ORDER BY PECNTGCONTRI DESC;
```

Answer:

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
ORDERID	TOTAL_AMOUNT	PECNTGCONTRI	
1001	57000	67.8992	
1005	9000	10.7209	
1009	8000	9.5297	
1002	4670	5.5630	
1010	1999	2.3812	
1006	1200	1.4295	
1004	789	0.9399	

Question-19

Write a SQL Query to find Top 2 Products by Sales within Each Category

Table Script:

```
CREATE TABLE ORDERS_TB2
```

```
(ORDERID INT, ORDERDATE DATE, CUSTOMER_NAME VARCHAR(20),PRODUCT_NAME  
VARCHAR(30), ORDER_QUANTITY TINYINT, TOTAL_AMOUNT INT, PRODUCT_CATEGORY  
VARCHAR(20));
```

```
INSERT INTO ORDERS_TB2 VALUES
```

```
(1001,"2025-01-15","Alice Smith","Laptop Pro",1,57000,"Electronics"),  
(1002,"2025-01-15","Bob Johnson","Ergonomic Chair",1,4670,"Furniture"),  
(1003,"2025-01-16","Charlie Brown","Wireless Mouse",2,430,"Electronics"),  
(1004,"2025-01-17","David Lee","Coffee Maker",3,789,"Home Appliances"),  
(1005,"2025-01-18","Alice Smith","External Hard Drive",1,9000,"Electronics"),  
(1006,"2025-01-18","Eve Davis","Bluetooth Speaker",4,1200,"Electronics"),  
(1007,"2025-01-19","Frank Green","Desk Lamp",1,300,"Home Decor"),  
(1008,"2025-01-20","Grace Hall","Gaming Keyboard",2,560,"Electronics"),  
(1009,"2025-01-20","Bob Johnson","Smart TV",1,8000,"Electronics"),  
(1010,"2025-01-21","Ivy White","Air Fryer",1,1999,"Home Appliances");
```

Solution:

```
SELECT * FROM (
SELECT PRODUCT_CATEGORY, PRODUCT_NAME,SUM(TOTAL_AMOUNT) AS SALES,
DENSE_RANK() OVER(PARTITION BY PRODUCT_CATEGORY ORDER BY SUM(TOTAL_AMOUNT)
DESC) AS RNK
FROM ORDERS_TB2
GROUP BY PRODUCT_CATEGORY, PRODUCT_NAME ) TBL
WHERE RNK <= 2;
```

Answer:

PRODUCT_CATEGORY	PRODUCT_NAME	SALES	RNK
Electronics	Laptop Pro	57000	1
Electronics	External Hard Drive	9000	2
Furniture	Ergonomic Chair	4670	1
Home Appliances	Air Fryer	1999	1
Home Appliances	Coffee Maker	789	2
Home Decor	Desk Lamp	300	1

Question-20

Write a SQL Query to find Customer Spend by Specific Categories (Cross-Tabulation)

Table Script:

```
CREATE TABLE ORDERS_TB2
```

```
(ORDERID INT, ORDERDATE DATE, CUSTOMER_NAME VARCHAR(20),PRODUCT_NAME  
VARCHAR(30), ORDER_QUANTITY TINYINT, TOTAL_AMOUNT INT, PRODUCT_CATEGORY  
VARCHAR(20));
```

```
INSERT INTO ORDERS_TB2 VALUES
```

```
(1001,"2025-01-15","Alice Smith","Laptop Pro",1,57000,"Electronics"),  
(1002,"2025-01-15","Bob Johnson","Ergonomic Chair",1,4670,"Furniture"),  
(1003,"2025-01-16","Charlie Brown","Wireless Mouse",2,430,"Electronics"),  
(1004,"2025-01-17","David Lee","Coffee Maker",3,789,"Home Appliances"),  
(1005,"2025-01-18","Alice Smith","External Hard Drive",1,9000,"Electronics"),  
(1006,"2025-01-18","Eve Davis","Bluetooth Speaker",4,1200,"Electronics"),  
(1007,"2025-01-19","Frank Green","Desk Lamp",1,300,"Home Decor"),  
(1008,"2025-01-20","Grace Hall","Gaming Keyboard",2,560,"Electronics"),  
(1009,"2025-01-20","Bob Johnson","Smart TV",1,8000,"Electronics"),  
(1010,"2025-01-21","Ivy White","Air Fryer",1,1999,"Home Appliances");
```

Solution:

```
SELECT CUSTOMER_NAME,  
SUM(CASE WHEN PRODUCT_CATEGORY = "ELECTRONICS" THEN TOTAL_AMOUNT ELSE 0  
END) AS ELECTRONINCS_SPEND,  
SUM(CASE WHEN PRODUCT_CATEGORY = "FURNITURE" THEN TOTAL_AMOUNT ELSE 0 END)  
AS FURNITURE_SPEND,  
SUM(CASE WHEN PRODUCT_CATEGORY = "HOME APPLIANCES" THEN TOTAL_AMOUNT ELSE  
0 END) HOMEAPPLI_SPEND,  
SUM(CASE WHEN PRODUCT_CATEGORY = "HOME DECOR" THEN TOTAL_AMOUNT ELSE 0  
END) AS HOMEDECOR_SPEND,  
sum(TOTAL_AMOUNT) AS TTL_SPEND  
FROM ORDERS_TB2  
GROUP BY CUSTOMER_NAME  
ORDER BY CUSTOMER_NAME;
```

Answer:

CUSTOMER_NAME	ELECTRONINCS_SPEND	FURNITURE_SPEND	HOMEAPPLI_SPEND	HOMEDECOR_SPEND	TTL_SPEND
Alice Smith	66000	0	0	0	66000
Bob Johnson	8000	4670	0	0	12670
Charlie Brown	430	0	0	0	430
David Lee	0	0	789	0	789
Eve Davis	1200	0	0	0	1200
Frank Green	0	0	0	300	300
Grace Hall	560	0	0	0	560
Ivy White	0	0	1999	0	1999

Question-21

Write a query to find the most expensive product in each category

Table Script:

```
CREATE TABLE PRODUCT_TB3
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(60), CATEGORY VARCHAR(60), PRICE FLOAT,
STOCK INT );
```

```
INSERT INTO PRODUCT_TB3 VALUES
(101,"Laptop","Electronics",1200,50),
(102,"Wireless Mouse","Accessories",25.5,200),
(103,"Coffee Maker","Home Appliances",85,75),
(104,"Smartphone","Electronics",899.99,120),
(105,"Desk Lamp","Home Goods",45.75,150),
(106,"Bluetooth Speaker","Accessories",60,90),
(107,"Smart TV","Electronics",1500,40),
(108,"Blender","Home Appliances",70.50,65),
(109,"Gaming Keyboard","Accessories",95.25,110),
(110,"Sofa","Furniture",550,25);
```

Solution:

```
SELECT PRODUCT_NAME,CATEGORY,PRICE FROM (
SELECT PRODUCT_NAME,CATEGORY,PRICE,
DENSE_RANK() OVER(PARTITION BY CATEGORY ORDER BY SUM(PRICE) DESC) AS RNK
FROM PRODUCT_TB3 GROUP BY PRODUCT_NAME,CATEGORY,PRICE) AS TBL
WHERE RNK <=1;
```

Answer:

Result Grid			
Filter Rows:			
Export:  Wrap Cell Content: 			
	PRODUCT_NAME	CATEGORY	PRICE
▶	Gaming Keyboard	Accessories	95.25
	Smart TV	Electronics	1500
	Sofa	Furniture	550
	Coffee Maker	Home Appliances	85
	Desk Lamp	Home Goods	45.75

Question-22

Write a query to list down all the products with a stock level below the average stock for their category.

Table Script:

```
CREATE TABLE PRODUCT_TB3
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(60), CATEGORY VARCHAR(60), PRICE FLOAT,
STOCK INT );
```

```
INSERT INTO PRODUCT_TB3 VALUES
(101,"Laptop","Electronics",1200,50),
(102,"Wireless Mouse","Accessories",25.5,200),
(103,"Coffee Maker","Home Appliances",85,75),
(104,"Smartphone","Electronics",899.99,120),
(105,"Desk Lamp","Home Goods",45.75,150),
(106,"Bluetooth Speaker","Accessories",60,90),
(107,"Smart TV","Electronics",1500,40),
(108,"Blender","Home Appliances",70.50,65),
(109,"Gaming Keyboard","Accessories",95.25,110),
(110,"Sofa","Furniture",550,25);
```

Solution:

```
SELECT * FROM (
SELECT PRODUCT_ID,PRODUCT_NAME,CATEGORY,STOCK,
AVG(STOCK) OVER(PARTITION BY CATEGORY) AS AVGSTOCK
FROM PRODUCT_TB3) TBL
WHERE STOCK < AVGSTOCK;
```

Answer:

Result Grid Filter Rows: Export: Wrap Cell Content:					
	PRODUCT_ID	PRODUCT_NAME	CATEGORY	STOCK	AVGSTOCK
▶	106	Bluetooth Speaker	Accessories	90	133.3333
	109	Gaming Keyboard	Accessories	110	133.3333
	101	Laptop	Electronics	50	70.0000
	107	Smart TV	Electronics	40	70.0000
	108	Blender	Home Appliances	65	70.0000

Question-23

Write a query to calculate the total value of the stock for each category

Table Script:

```
CREATE TABLE PRODUCT_TB3
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(60), CATEGORY VARCHAR(60), PRICE FLOAT,
STOCK INT );
```

```
INSERT INTO PRODUCT_TB3 VALUES
(101,"Laptop","Electronics",1200,50),
(102,"Wireless Mouse","Accessories",25.5,200),
(103,"Coffee Maker","Home Appliances",85,75),
(104,"Smartphone","Electronics",899.99,120),
(105,"Desk Lamp","Home Goods",45.75,150),
(106,"Bluetooth Speaker","Accessories",60,90),
(107,"Smart TV","Electronics",1500,40),
(108,"Blender","Home Appliances",70.50,65),
(109,"Gaming Keyboard","Accessories",95.25,110),
(110,"Sofa","Furniture",550,25);
```

Solution:

```
SELECT CATEGORY,SUM(STOCK * PRICE) AS TTLSTOCKVALUE
FROM PRODUCT_TB3
GROUP BY CATEGORY;
```

Answer:

Result Grid Filter Rows: Export: Wrap Cell Content:		
	CATEGORY	TTLSTOCKVALUE
▶	Electronics	227998.798828125
	Accessories	20977.5
	Home Appliances	10957.5
	Home Goods	6862.5
	Furniture	13750

Question-24

Write a query to find products with a price higher than the overall average price of all products.

Table Script:

```
CREATE TABLE PRODUCT_TB3  
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(60), CATEGORY VARCHAR(60), PRICE FLOAT,  
STOCK INT );
```

```
INSERT INTO PRODUCT_TB3 VALUES  
(101,"Laptop","Electronics",1200,50),  
(102,"Wireless Mouse","Accessories",25.5,200),  
(103,"Coffee Maker","Home Appliances",85,75),  
(104,"Smartphone","Electronics",899.99,120),  
(105,"Desk Lamp","Home Goods",45.75,150),  
(106,"Bluetooth Speaker","Accessories",60,90),  
(107,"Smart TV","Electronics",1500,40),  
(108,"Blender","Home Appliances",70.50,65),  
(109,"Gaming Keyboard","Accessories",95.25,110),  
(110,"Sofa","Furniture",550,25);
```

Solution:

```
SELECT * FROM (  
SELECT PRODUCT_NAME,PRICE,AVG(PRICE) OVER() AS AVGPRICE  
FROM PRODUCT_TB3 ) AS TBL WHERE PRICE >=AVGPRICE;
```

Answer:

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
PRODUCT_NAME	PRICE	AVGPRICE	
Laptop	1200	453.1989990234375	
Smartphone	899.99	453.1989990234375	
Smart TV	1500	453.1989990234375	
Sofa	550	453.1989990234375	

Question-25

Write a query to find categories that have more than 2 products.

Table Script:

```
CREATE TABLE PRODUCT_TB3  
(PRODUCT_ID INT, PRODUCT_NAME VARCHAR(60), CATEGORY VARCHAR(60), PRICE FLOAT,  
STOCK INT );
```

```
INSERT INTO PRODUCT_TB3 VALUES  
(101,"Laptop","Electronics",1200,50),  
(102,"Wireless Mouse","Accessories",25.5,200),  
(103,"Coffee Maker","Home Appliances",85,75),  
(104,"Smartphone","Electronics",899.99,120),  
(105,"Desk Lamp","Home Goods",45.75,150),  
(106,"Bluetooth Speaker","Accessories",60,90),  
(107,"Smart TV","Electronics",1500,40),  
(108,"Blender","Home Appliances",70.50,65),  
(109,"Gaming Keyboard","Accessories",95.25,110),  
(110,"Sofa","Furniture",550,25);
```

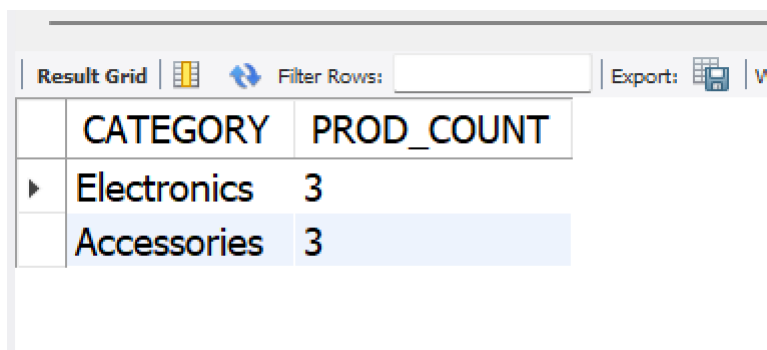
Solution:

```
SELECT CATEGORY,COUNT(PRODUCT_NAME) AS PROD_COUNT  
FROM PRODUCT_TB3
```

```
GROUP BY CATEGORY
```

```
HAVING PROD_COUNT > 2
```

Answer:



The screenshot shows a database query result grid. At the top, there are tabs for 'Result Grid', 'Filter Rows', and 'Export'. The 'Result Grid' tab is active. Below the tabs, there is a table with two columns: 'CATEGORY' and 'PROD_COUNT'. The table contains two rows: 'Electronics' with a count of 3, and 'Accessories' with a count of 3. The 'Accessories' row is highlighted in blue.

CATEGORY	PROD_COUNT
Electronics	3
Accessories	3

Question-26

Write a Query Calculate the total revenue for each month and the percentage growth compared to the previous month.

Table Script:

```
CREATE TABLE ORDERS_TB3
(ORDER_ID INT, USER_ID INT, ORDER_DATE DATE, TOTAL_AMOUNT INT, ORDER_STATUS
VARCHAR(60));
```

```
INSERT INTO ORDERS_TB3 VALUES
(101,1,"2025-01-15",150,"Completed"),
(102,2,"2025-01-20",200,"Completed"),
(103,1,"2025-02-10",120,"Completed"),
(104,3,"2025-02-15",320,"Completed"),
(105,2,"2025-03-05",250,"Completed");
```

Solution:

```
SELECT MNTH,TTLREVENUE,
(((TTLREVENUE-PREVMNTHREVENUE) / PREVMNTHREVENUE) * 100) AS GROWTHPCT
FROM(SELECT *, LAG(TTLREVENUE,1) OVER(ORDER BY MNTH) AS PREVMNTHREVENUE
FROM(SELECT MONTH(ORDER_DATE) AS MNTH,SUM(TOTAL_AMOUNT) AS TTLREVENUE
FROM ORDERS_TB3 GROUP BY MNTH)A ) B
```

Answer:

Result Grid			
		Filter Rows:	
		Export:	Wrap Cell Content
	MNTH	TTLREVENUE	GROWTHPCT
▶	1	350	NULL
	2	440	25.71
	3	250	-43.18

Question-27

write a query to get top 2 users by total spend for each month

Table Script:

```
CREATE TABLE ORDERS_TB3
(ORDER_ID INT, USER_ID INT, ORDER_DATE DATE, TOTAL_AMOUNT INT, ORDER_STATUS
VARCHAR(60));
```

```
INSERT INTO ORDERS_TB3 VALUES
(101,1,"2025-01-15",150,"Completed"),
(102,2,"2025-01-20",200,"Completed"),
(103,1,"2025-02-10",120,"Completed"),
(104,3,"2025-02-15",320,"Completed"),
(105,2,"2025-03-05",250,"Completed");
```

Solution:

```
SELECT * FROM
(SELECT USER_ID,MNTH, TTLSPEND,
DENSE_RANK() OVER(PARTITION BY MNTH ORDER BY TTLSPEND DESC) AS RNK
FROM
(SELECT USER_ID,MONTH(ORDER_DATE) AS MNTH, SUM(TOTAL_AMOUNT) AS TTLSPEND
FROM ORDERS_TB3
GROUP BY USER_ID,MNTH) A) B
WHERE RNK <=2;
```

Answer:

Result Grid				
Filter Rows:		Export:		
Wrap Cell Content:				
	USER_ID	MNTH	TTLSPEND	RNK
▶	2	1	200	1
	1	1	150	2
	3	2	320	1
	1	2	120	2
	2	3	250	1

Question-28

Write a query to find all products where the product_code starts with the letters 'ABC'.

Table Script:

```
CREATE TABLE PRODUCT_TB4  
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN  
VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES  
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),  
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),  
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),  
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),  
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),  
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT PRODUCT_CODE, PRODUCT_NAME FROM PRODUCT_TB4  
WHERE PRODUCT_CODE LIKE 'ABC%';
```

Answer:

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	PRODUCT_CODE	PRODUCT_NAME		
▶	ABC-123-X	Super-Fast Laptop		
	ABC-456-A	Smart Watch Series 5		
	ABC-789-X	Gaming Keyboard		

Question-29

Write a query to list all products where the product_code ends with the letter 'X'?

Table Script:

```
CREATE TABLE PRODUCT_TB4  
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN  
VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES  
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),  
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),  
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),  
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),  
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),  
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT PRODUCT_CODE, PRODUCT_NAME FROM PRODUCT_TB4  
WHERE PRODUCT_CODE LIKE '%X';
```

Answer:

483

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
PRODUCT_CODE	PRODUCT_NAME			
ABC-123-X	Super-Fast Laptop			
ABC-789-X	Gaming Keyboard			

Question-30

Write a query to find any product that has the word 'Watch' anywhere in its product_name?

Table Script:

```
CREATE TABLE PRODUCT_TB4  
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN  
VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES  
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),  
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),  
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),  
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),  
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),  
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT PRODUCT_NAME FROM PRODUCT_TB4  
WHERE PRODUCT_NAME LIKE "%WATCH%";
```

Answer:

Result Grid	Filter Rows:	Export:	Wrap Cell Co
PRODUCT_NAME			
▶ Smart Watch Series 5			

Question-31

write a query to identify all items stored in bins that start with 'A', followed by any single character, followed by a dash '-'

Table Script:

```
CREATE TABLE PRODUCT_TB4
```

```
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES
```

```
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),  
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),  
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),  
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),  
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),  
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT * FROM PRODUCT_TB4
```

```
WHERE STORAGE_BIN LIKE 'A_-%';
```

Answer:

ITEM_ID	PRODUCT_CODE	PRODUCT_NAME	STORAGE_BIN	LAST_AUDITED
1	ABC-123-X	Super-Fast Laptop	A1-01	2025-01-10
3	ABC-456-A	Smart Watch Series 5	A1-02	2025-01-20
5	ABC-789-X	Gaming Keyboard	A2-01	2025-02-10

Question-32

write a query to find all product_code values where the second character is strictly the letter 'B'.

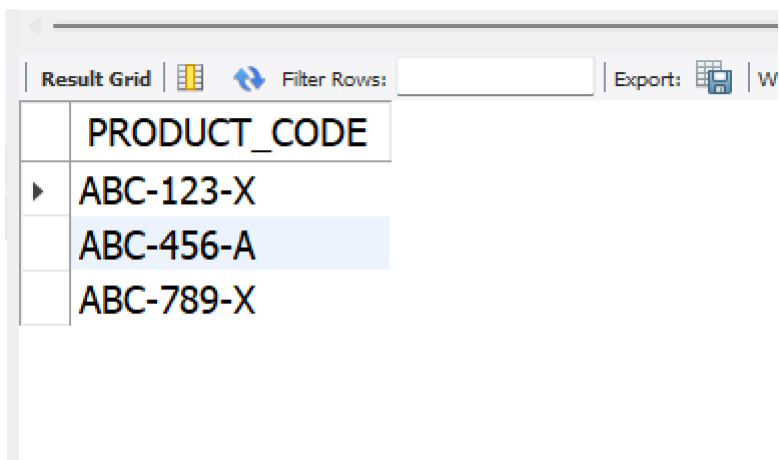
Table Script:

```
CREATE TABLE PRODUCT_TB4  
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN  
VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES  
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),  
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),  
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),  
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),  
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),  
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT PRODUCT_CODE FROM PRODUCT_TB4  
WHERE PRODUCT_CODE LIKE "_B%";
```

Answer:

	PRODUCT_CODE
▶	ABC-123-X
	ABC-456-A
	ABC-789-X

Question-33

write a query to find all products containing the word 'laptop', even if it's written as 'Laptop' or 'LAPTOP' in the database.

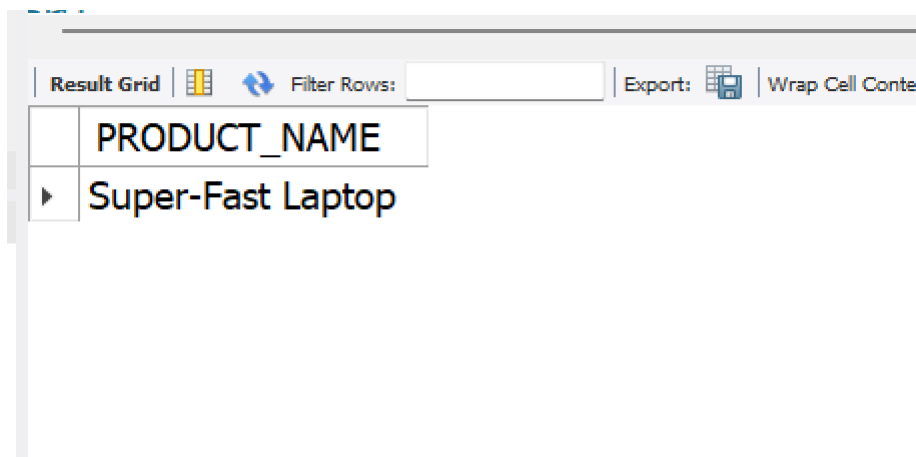
Table Script:

```
CREATE TABLE PRODUCT_TB4  
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN  
VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES  
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),  
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),  
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),  
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),  
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),  
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT PRODUCT_NAME FROM  
PRODUCT_TB4  
WHERE PRODUCT_NAME LIKE "%laptop%";
```

Answer:

The screenshot shows a database query result grid. The grid has a header row with the column name 'PRODUCT_NAME'. Below the header, there is one data row containing the value 'Super-Fast Laptop'. The grid is displayed in a window with a toolbar at the top containing options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Conte'.

PRODUCT_NAME
Super-Fast Laptop

Question-34

write a query to retrieve all items except those located in storage bins starting with the letter 'B'.

Table Script:

```
CREATE TABLE PRODUCT_TB4
(ITEM_ID INT, PRODUCT_CODE VARCHAR(60), PRODUCT_NAME VARCHAR(60), STORAGE_BIN
VARCHAR(60), LAST_AUDITED DATE);
```

```
INSERT INTO PRODUCT_TB4 VALUES
(1,"ABC-123-X", "Super-Fast Laptop","A1-01","2025-01-10"),
(2,"XYZ-999-B", "Blue Mouse (V2)","B2-05","2025-02-15"),
(3,"ABC-456-A", "Smart Watch Series 5","A1-02","2025-01-20"),
(4,"DEF-111-Z", "Office Chair","C1-10","2025-02-01"),
(5,"GHI-222-Y", "Portable HDD 1TB","B2-08","2025-01-05"),
(5,"ABC-789-X", "Gaming Keyboard","A2-01","2025-02-10");
```

Solution:

```
SELECT * FROM PRODUCT_TB4
WHERE STORAGE_BIN NOT LIKE "B%";
```

Answer:

Result Grid					
Filter Rows:		Export:		Wrap Cell Content:	
	ITEM_ID	PRODUCT_CODE	PRODUCT_NAME	STORAGE_BIN	LAST_AUDITED
▶	1	ABC-123-X	Super-Fast Laptop	A1-01	2025-01-10
	3	ABC-456-A	Smart Watch Series 5	A1-02	2025-01-20
	4	DEF-111-Z	Office Chair	C1-10	2025-02-01
	5	ABC-789-X	Gaming Keyboard	A2-01	2025-02-10

Question-35

Write a query to list all bookings showing the Guest Name, Room Number, and Total Bill.

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
SELECT CONCAT(G.FIRST_NAME,' ',G.LAST_NAME) as GUESTNAME, B.TOTAL_BILL,
R.ROOM_NUMBER
FROM GUEST_TB G INNER JOIN BOOKINGS_TB B INNER JOIN ROOMS_TB R
ON G.GUEST_ID = B.GUEST_ID AND B.ROOM_ID = R.ROOM_ID;
```

Answer:

Result Grid			
Filter Rows:			
Export: Wrap Cell Contents:			
	GUESTNAME	TOTAL_BILL	ROOM_NUMBER
▶	PRIYA SHARMA	300	101
	RAJESH KUMAR	1400	102
	RAJESH KUMAR	150	103
	AMIT SINGH	4000	104

Question-36

Write a query to find the names of all guests who stayed in a 'Suite' room type.

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
SELECT CONCAT(G.FIRST_NAME,' ',G.LAST_NAME)AS GUESTNAME
FROM GUEST_TB G INNER JOIN BOOKINGS_TB B INNER JOIN ROOMS_TB R ON G.GUEST_ID =
B.GUEST_ID AND B.ROOM_ID = R.ROOM_ID WHERE R.ROOM_TYPE = "SUITE";
```

Answer:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	GUESTNAME
▶	RAJESH KUMAR

Question-37

write a query to identify any guests who have never made a booking.

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
SELECT CONCAT(G.FIRST_NAME,' ',G.LAST_NAME) AS GUESTNAME
FROM GUEST_TB G LEFT JOIN BOOKINGS_TB B
ON G.GUEST_ID = B.GUEST_ID
WHERE B.GUEST_ID IS NULL;
```

Answer:

Result Grid	Filter Rows:	Exp
GUESTNAME		
SNEHA PATIL		

Question-38

Write a query to find which Room Type has generated the highest total revenue for the hotel?

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
WITH CTE AS(
SELECT R.ROOM_TYPE,SUM(B.TOTAL_BILL) AS REVENUE,
DENSE_RANK() OVER(ORDER BY SUM(B.TOTAL_BILL) DESC) AS RNK
FROM ROOMS_TB R INNER JOIN BOOKINGS_TB B
ON R.ROOM_ID = B.ROOM_ID
GROUP BY R.ROOM_TYPE)
SELECT * FROM CTE WHERE RNK = 1;
```

Answer:

Result Grid			
Filter Rows:		Export:	
	ROOM_TYPE	REVENUE	RNK
▶	PENTHOUSE	4000	1

Question-39

Write a query to calculate the average total_bill paid by Loyalty Members versus Non-Loyalty Members.

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
SELECT G.LOYALTY_MEMBER,AVG(B.TOTAL_BILL) AS AVGBILL
FROM GUEST_TB G INNER JOIN BOOKINGS_TB B
ON G.GUEST_ID = B.GUEST_ID
GROUP BY G.LOYALTY_MEMBER;
```

Answer:

Result Grid		
Filter Rows:		
Export: Excel CSV PDF		
Wrap Cell Content: On Off		
LOYALTY_MEMBER	AVGBILL	
Yes	1850.0000	
No	300.0000	

Question-40

Write a query to find all bookings where the guest stayed for more than 3 nights.

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
SELECT *,DATEDIFF(CHECK_OUT_DATE,CHECK_IN_DATE) AS NIGHTS
FROM BOOKINGS_TB
WHERE DATEDIFF(CHECK_OUT_DATE,CHECK_IN_DATE) > 3;
```

Answer:

BOOKING_ID	GUEST_ID	ROOM_ID	CHECK_IN_DATE	CHECK_OUT_DATE	TOTAL_BILL	NIGHTS
BOO1	1	12	2025-02-01	2025-02-05	1400	4
BOO4	3	14	2025-02-20	2025-02-25	4000	5

Question-41

Write a query to find all guests whose email domain is '@ndac.com' and list the room numbers they booked.

Table Script:

```
CREATE TABLE GUEST_TB
(GUEST_ID INT, FIRST_NAME VARCHAR(60), LAST_NAME VARCHAR(60), EMAIL
VARCHAR(60), LOYALTY_MEMBER VARCHAR(20));
INSERT INTO GUEST_TB VALUES
(1,"RAJESH","KUMAR","rajesh@yahoo.com","Yes"),
(2,"PRIYA","SHARMA","priya@gmail.com","No"),
(3,"AMIT","SINGH","amit.s@outlook.com","Yes"),
(4,"SNEHA","PATIL","sneha.p@ndac.com","No");

CREATE TABLE ROOMS_TB
(ROOM_ID INT, ROOM_NUMBER INT, ROOM_TYPE VARCHAR(30), PRICE_PER_NIGHT INT, STATUS
VARCHAR(40));
INSERT INTO ROOMS_TB VALUES
(11,101,"DELUXE",150,"AVAILABLE"),
(12,102,"SUITE",350,"OCCUPIED"),
(13,103,"DELUXE",150,"AVAILABLE"),
(14,104,"PENTHOUSE",800,"OCCUPIED");

CREATE TABLE BOOKINGS_TB
(BOOKING_ID VARCHAR(30), GUEST_ID INT, ROOM_ID INT, CHECK_IN_DATE
DATE, CHECK_OUT_DATE DATE, TOTAL_BILL INT);
INSERT INTO BOOKINGS_TB VALUES
("BOO1",1,12,"2025-02-01","2025-02-05",1400),
("BOO2",2,11,"2025-02-10","2025-02-12",300),
("BOO3",1,13,"2025-02-15","2025-02-16",150),
("BOO4",3,14,"2025-02-20","2025-02-25",4000);
```

Solution:

```
SELECT CONCAT(G.FIRST_NAME,' ',G.LAST_NAME) AS GUESTNAME, R.ROOM_NUMBER
FROM GUEST_TB G
LEFT JOIN BOOKINGS_TB B ON G.GUEST_ID = B.GUEST_ID
LEFT JOIN ROOMS_TB R ON B.ROOM_ID = R.ROOM_ID
WHERE G.EMAIL LIKE "%@ndac.com%";
```

Answer:

Result Grid		Filter Rows:	Export:
GUESTNAME	ROOM_NUMBER		
SNEHA PATIL	NULL		

Question-42

Write a query for each City, calculate the total number of trips, the total distance covered, and the average fare per trip.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT V.CITY,COUNT(T.TRIP_ID) AS TRIPCPUNT,SUM(T.DISTANCE_KM) AS
TTLDISTANCE,AVG(T.FARE_AMOUNT) AS AVGFARE
FROM VEHICLES_TB V JOIN TRIPS_TB T
ON V.VEHICLE_ID = T.VEHICLE_ID GROUP BY V.CITY;
```

Answer:

	CITY	TRIPCPUNT	TTLDISTANCE	AVGFARE
▶	Singapore	9	162.19999980926514	46.3333
	Mumbai	2	14.5	22.5000
	Bangkok	1	18.5	45.0000

Question-43

Write a query to Identify Vehicles that have completed more than 3 trips and have generated a total revenue (fare_amount) greater than or equals to \$100.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT V.MODEL,COUNT(T.TRIP_ID) AS TRIPCOUN, SUM(T.FARE_AMOUNT) AS TTLREVENUE
FROM VEHICLES_TB V JOIN TRIPS_TB T
ON V.VEHICLE_ID = T.VEHICLE_ID
GROUP BY V.MODEL HAVING TRIPCOUN >= 3 AND TTLREVENUE > 100;
```

Answer:

MODEL	TRIPCOUN	TTLREVENUE
Tesla Model 3	4	107
Toyota Prius	3	155

Question-44

Write a query to find all trips where the fare_amount is higher than the average fare of all trips in the database.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT * FROM TRIPS_TB WHERE FARE_AMOUNT
>(SELECT AVG(FARE_AMOUNT) FROM TRIPS_TB);
```

Answer:

TRIP_ID	VEHICLE_ID	GUEST_ID	TRIP_DATE	DURATION	DISTANCE_KM	FARE_AMOUNT
T102	V002	46054	2026-02-01	45	22	55
T104	V004	46055	2026-02-02	60	35.5	85
T107	V006	46057	2026-02-04	40	18.5	45
T109	V004	46058	2026-02-05	50	28	70
T112	V002	46059	2026-02-06	55	25	60

Question-45

Write a query using CTE to calculate the total revenue per Vehicle Category and then query that CTE to find only the categories that contribute more than 50% of the total company revenue.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));

INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");

CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));

INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
WITH TBL AS(
SELECT V.CATEGORY,SUM(T.FARE_AMOUNT) AS TTLREVENUE
FROM VEHICLES_TB V JOIN TRIPS_TB T
ON V.VEHICLE_ID = T.VEHICLE_ID GROUP BY V.CATEGORY)
SELECT CATEGORY, TTLREVENUE FROM TBL
WHERE TTLREVENUE > (SELECT SUM(TTLREVENUE) / 2 FROM TBL);
```

Answer:

	CATEGORY	TTLREVENUE
▶	Electric	307

Question-46

Write a query to list all vehicles in Mumbai, showing their model and the trip_id of any trips they have taken. If a vehicle has never taken a trip (like V003), show 'No Trips' for the trip_id.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT V.MODEL,CASE WHEN T.TRIP_ID IS NULL THEN "No Trips" ELSE T.TRIP_ID END AS
TRIP_ID FROM VEHICLES_TB V LEFT JOIN TRIPS_TB T
ON V.VEHICLE_ID = T.VEHICLE_ID WHERE V.CITY = "Mumbai";
```

Answer:

Result Grid		Filter Rows:
	MODEL	TRIP_ID
▶	BMW i4	No Trips
	Honda Civic	T105
	Honda Civic	T110

Question-47

Write a query for each city, rank the vehicles based on the total distance they have traveled, from highest to lowest.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT V.CITY,V.VEHICLE_ID,SUM(T.DISTANCE_KM) AS DIST,
DENSE_RANK() OVER(PARTITION BY V.CITY ORDER BY SUM(T.DISTANCE_KM) DESC ) AS RNK
FROM VEHICLES_TB V JOIN TRIPS_TB T ON V.VEHICLE_ID = T.VEHICLE_ID GROUP BY
V.CITY,V.VEHICLE_ID;
```

Answer:

CITY	VEHICLE_ID	DIST	RNK
Bangkok	V006	18.5	1
Mumbai	V005	14.5	1
Singapore	V004	63.5	1
Singapore	V002	62	2
Singapore	V001	36.69999980926514	3

Question-48

#Write a query to Find all trips taken by vehicles where the model name contains the word 'Model'

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT V.MODEL,T.TRIP_ID FROM VEHICLES_TB V JOIN TRIPS_TB T
ON V.VEHICLE_ID = T.VEHICLE_ID
WHERE V.MODEL LIKE "%Model%";
```

Answer:

	MODEL	TRIP_ID
▶	Tesla Model 3	T101
	Tesla Model 3	T103
	Tesla Model 3	T108
	Tesla Model 3	T111

Question-49

Write a query to calculate the "Efficiency" (Distance per Minute) for every trip. Ensure the result is a decimal (float) and not truncated. Compare the average efficiency of the 'Electric' category vs. the 'Petrol' category.

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
SELECT V.CATEGORY,AVG(EFFICIENCY) FROM(
SELECT VEHICLE_ID, (DISTANCE_KM/DURATION) AS EFFICIENCY
FROM TRIPS_TB) T JOIN VEHICLES_TB V
ON V.VEHICLE_ID = T.VEHICLE_ID
WHERE V.CATEGORY IN ("Electric","Petrol")
GROUP BY V.CATEGORY;
```

Answer:

CATEGORY	AVG(EFFICIENCY)
Electric	0.459710882537219
Petrol	0.4166666666666667

Question-50

Use a CTE for this query to calculate the average efficiency per vehicle first. Then, in the main query, find which City has the highest overall "Electric Efficiency."

Table Script:

```
CREATE TABLE VEHICLES_TB
(VEHICLE_ID VARCHAR(60) PRIMARY KEY,MODEL VARCHAR(60),CATEGORY VARCHAR(60),CITY
VARCHAR(60), STATUS VARCHAR(60));
```

```
INSERT INTO VEHICLES_TB VALUES
("V001","Tesla Model 3","Electric","Singapore","Active"),
("V002","Toyota Prius","Hybrid","Singapore","Active"),
("V003","BMW i4","Electric","Mumbai","Maintenance"),
("V004","Mercedes EQE","Electric","Singapore","Active"),
("V005","Honda Civic","Petrol","Mumbai","Active"),
("V006","BYD Atto 3","Electric","Bangkok","Active");
```

```
CREATE TABLE TRIPS_TB
(TRIP_ID VARCHAR(60) PRIMARY KEY,VEHICLE_ID VARCHAR(60), GUEST_ID INT,TRIP_DATE
DATE, DURATION INT,
DISTANCE_KM FLOAT,FARE_AMOUNT INT,FOREIGN KEY (VEHICLE_ID) REFERENCES
VEHICLES_TB (VEHICLE_ID));
```

```
INSERT INTO TRIPS_TB VALUES
("T101","V001",46054,"2026-02-01",25,12.5,35),
("T102","V002",46054,"2026-02-01",45,22,55),
("T103","V001",46055,"2026-02-02",15,5.2,18),
("T104","V004",46055,"2026-02-02",60,35.5,85),
("T105","V005",46056,"2026-02-03",20,8,25),
("T106","V002",46056,"2026-02-03",30,15,40),
("T107","V006",46057,"2026-02-04",40,18.5,45),
("T108","V001",46057,"2026-02-04",10,3,12),
("T109","V004",46058,"2026-02-05",50,28,70),
("T110","V005",46058,"2026-02-05",15,6.5,20),
("T111","V001",46059,"2026-02-06",35,16,42),
("T112","V002",46059,"2026-02-06",55,25,60);
```

Solution:

```
WITH VEHICLE_EFFICIENCY AS (
SELECT V.CITY,V.CATEGORY,V.VEHICLE_ID,SUM(T.DISTANCE_KM/T.DURATION) AS EFFICIENCY
FROM VEHICLES_TB V JOIN TRIPS_TB T
ON V.VEHICLE_ID = T.VEHICLE_ID GROUP BY V.CITY,V.CATEGORY,V.VEHICLE_ID
),CITY_EFFICIENCY AS(SELECT CITY,CATEGORY,AVG(EFFICIENCY) AS AVGEFFICIENCY
FROM VEHICLE_EFFICIENCY WHERE CATEGORY = "Electric" GROUP BY CITY,CATEGORY)
SELECT CITY,AVGEFFICIENCY FROM CITY_EFFICIENCY ORDER BY AVGEFFICIENCY DESC LIMIT 1;
```

Answer:

Result Grid		Filter Rows:	Export:
CITY	AVGEFFICIENCY		
▶ Singapore	1.3777380888802666		